# Nexivarin 10 mg Tablets — eCTD Submission

Module 4 — Nonclinical Study Reports

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| **Submission Type** | NDA — 505(b)(1) | **Sequence Number** | 0000 |
| **Submission Date** | 2021-03-15 | **Section** | 4.2.3.2 |
| **Prepared by** | PharmaCorp Inc. | **Confidentiality** | Proprietary and Confidential |

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**PharmaCorp Inc.**
NDA 123456

# Nexivarin 10 mg Tablets

## Section 4.2.3.2
## Repeat-Dose Toxicity

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# 4.2.3.2 Repeat-Dose Toxicity Studies

## Study 4.2.3.2-001: 26-Week Oral Gavage Toxicity Study in Rats

| Parameter | Value |
|---|---|
| Species/Strain | Sprague-Dawley rat (Crl:CD(SD)), male and female |
| Number/Group | 15/sex/group (5/sex/group satellite TK) |
| Route | Oral gavage (0.5% methylcellulose) |
| Dose Levels | 0, 3, 10, 30, 100 mg/kg/day |
| Duration | 26 weeks + 4-week recovery (control and high-dose groups) |
| GLP Status | GLP Compliant (US 21 CFR Part 58) |
| Study Number | NXV-TOX-006 |

### *Key Findings*

| Parameter | 0 mg/kg | 3 mg/kg | 10 mg/kg | 30 mg/kg | 100 mg/kg |
|---|---|---|---|---|---|
| Body weight change | ↔ | ↔ | ↔ | ↓3% M/F (NS) | ↓8% M/F ($p<0.05$) |
| Food consumption | ↔ | ↔ | ↔ | ↔ | ↓5% (NS) |
| Clinical observations | Normal | Normal | Normal | Mild facial flushing (F) | Peripheral edema, flushing |
| Haematology | Normal | Normal | Normal | Normal | Normal |
| Clinical chemistry | Normal | Normal | Normal | Normal | Mild ALT ↑ (≤2× ULN) |
| Organ weights — Heart | 100% | 100% | 101% | 102% | 103% (NS) |
| Organ weights — Liver | 100% | 100% | 100% | 102% | 108% ($p<0.05$) |
| Histopathology | Normal | Normal | Normal | Arteriolar dilation (trace, F) | Arteriolar dilation (mild, M+F); hepatocyte vacuolation (minimal) |
| TK $AUC_{0-24}$ (ng·h/mL) | 0 | 42±8 | 156±22 | 498±64 | 1,842±312 |
| Safety margin vs. clinical AUC (412 ng·h/mL) | — | 0.1× | 0.4× | 1.2× | 4.5× |

**NOAEL: 30 mg/kg/day** (AUC safety margin: 1.2-fold at NOAEL). Findings at 100 mg/kg were characterized by pharmacologically predictable hemodynamic effects (edema, flushing) and minimal adaptive hepatic changes. All findings at 100 mg/kg were fully or partially reversible during the 4-week recovery period.

## Study 4.2.3.2-002: 39-Week Oral Toxicity Study in Dogs

| Parameter | Value |
|---|---|
| Species/Strain | Beagle dog (Canis familiaris), male and female |
| Number/Group | 4/sex/group |
| Route | Oral (gelatin capsule) |
| Dose Levels | 0, 1, 3, 10, 30 mg/kg/day |
| Duration | 39 weeks + 4-week recovery (control and high-dose) |
| GLP Status | GLP Compliant |
| Study Number | NXV-TOX-009 |

| Finding | 0 mg/kg | 3 mg/kg | 10 mg/kg | 30 mg/kg |
|---|---|---|---|---|
| Clinical signs | Normal | Normal | Mild peripheral edema (1/4F) | Peripheral edema (3/4M, 4/4F); flushing |
| Body weight | ↔ | ↔ | ↔ | ↓5% (NS) |
| ECG (QTc) | Normal | Normal | Normal | Normal (no prolongation) |
| Blood pressure (24h telemetry) | Baseline | ↓5 mmHg (NS) | ↓12 mmHg ($p<0.05$) | ↓22 mmHg ($p<0.01$) |
| Histopathology | Normal | Normal | Arteriolar dilation (minimal) | Arteriolar dilation (mild); gingival hyperplasia (1/4M, minimal) |
| TK AUC0-24 (ng·h/mL) | 0 | 384±54 | 1,248±188 | 4,120±542 |
| Safety margin vs. clinical AUC | — | 0.9× | 3.0× | 10.0× |

**NOAEL: 10 mg/kg/day** (safety margin: 3.0-fold). Findings at 30 mg/kg/day were pharmacologically mediated (hemodynamic) and consistent with calcium channel blockade. Gingival hyperplasia, a class effect of calcium channel blockers, was observed at the high dose only.